

PRODUCT 01 - E-Commerce D2C Sales OS - Developer Build Sheet

Stack: Frontend Next.js + Backend Node.js + MongoDB

Backend: npm i express mongoose cors dotenv jsonwebtoken bcryptjs multer razorpay shiprocket tally-connector node-cron axios nodemailer

Frontend: npx create-next-app@latest frontend --tailwind --app, npm i axios razorpay

1. Setup

Backend:

cd backend

npm init -y

npm i express mongoose cors dotenv jsonwebtoken bcryptjs multer nodemailer node-cron exceljs razorpay axios

Frontend:

npx create-next-app@latest frontend --tailwind --app

cd frontend

npm i axios

2. Structure

backend/

app.js, cron.js, seed.js

models/Company.js, Product.js, Order.js, TallySync.js, AbandonedCart.js

routes/auth.js, products.js, orders.js, tally.js, whatsapp.js, payment.js, audit.js

frontend/

app/login/page.jsx, dashboard/page.jsx, products/page.jsx, orders/page.jsx, tally/page.jsx,

abandoned/page.jsx, settings/page.jsx

lib/api.js

3. Models

Company: _id, name, subdomain unique, ownerEmail, passwordHash, createdAt

Product: _id, companyId ref Company, title, sku unique, price Number, gstPercent, stock Number, images [fileUrl], tallyItemId, status active/inactive

Order: _id, companyId, customerName, phone, items [{productId, qty, price}], total, gstTotal, paymentMode COD/UPI/Razorpay, paymentStatus pending/paid/failed, rtoScore 0-100, shiprocketAwb, status valid/packed/shipped/delivered/rto, createdAt

TallySync: companyId, lastSyncAt Date, itemsSynced Number, status success/failed

AbandonedCart: companyId, phone, productId ref Product, cartValue, reminderDays [0,1,3], recovered bool, createdAt

4. Backend APIs

POST /api/auth/register - name, subdomain, ownerEmail, password -> hash -> create Company -> JWT

POST /api/auth/login - subdomain, email, password -> JWT companyId

POST /api/products/create - Bearer, multer multiple images 5 max, title, sku, price, gstPercent, stock -> save /uploads/, fileUrl, create Product

GET /api/products/list?companyId=&status= - sort createdAt desc

POST /api/orders/create - companyId, phone, items, paymentMode -> total+GST, rtoScore, create Order

GET /api/orders/list?companyId=&status= - sort expiry asc

GET /api/orders/stats?companyId= - total, rtoCount, deliveredCount

POST /api/orders/abandoned/check - companyId, phone, productId, cartValue -> create AbandonedCart

POST /api/payment/verify - razorpay_payment_id, orderId -> verify, update paid

POST /api/tally/sync - companyId, tallyData -> sync stock, TallySync log

POST /api/whatsapp/abandoned - abandonedCartId -> send WhatsApp image + UPI link via Meta API

GET /api/audit/export?companyId=&year=2026 - Excel Title, SKU, Total, GST, Status

Cron: 0 8,15,20 * * * daily -> loop AbandonedCart recovered false, diffDays = now-createdAt, if reminderDays includes diffDays -> WhatsApp + SMS + log

5. Frontend Pages

lib/api.js: axios baseURL http://localhost:5000/api, Bearer token

login: subdomain, email, password -> POST /auth/login -> save token, companyId -> push dashboard

dashboard: Cards Total Products, Total Orders, Abandoned Count, RTO %, table RTO Risk Orders from /orders/list?status=rto

products: Form Title, SKU, Price, GST %, Stock, File multiple, Upload -> FormData POST
/products/create, table with image thumbnail
orders: Table orders phone, total, paymentMode, status, button Mark Shipped -> Shiprocket API,
Send Invoice WhatsApp
abandoned: Table AbandonedCart phone, cartValue, recovered bool, button Send WhatsApp Now
settings: Toggles WhatsApp, SMS, UPI AutoPay, Language English/Marathi/Hindi, Razorpay Key,
Shiprocket Email/Pass, Test Alert

6. .env

```
MONGO_URI=mongodb://localhost:27017/ecommerce_os
JWT_SECRET=your_jwt_secret_123
RAZORPAY_KEY=rzp_test_xxx
RAZORPAY_SECRET=xxx
SHIPROCKET_EMAIL=xxx
SHIPROCKET_PASS=xxx
WHATSAPP_TOKEN=EAxxxx
WHATSAPP_PHONE_ID=123xxx
FAST2SMS_KEY=xxx
EMAIL_USER=gmail
EMAIL_PASS=app_password
ALERT_TO=owner@company.com
```

7. Run + Done Checklist

Run: node app.js starts cron, npm run dev

Done:

- Company register/login with subdomain works
- Product upload with GST calculated works
- Product list filtered by companyId works
- Dashboard cards correct
- Order create with RTO score works
- Abandoned cron sends WhatsApp image + UPI at 0,1,3 days
- Tally sync updates stock
- Razorpay verify updates paid
- Excel export downloads
- WhatsApp Marathi/Hindi toggle works